Author Response: TriPoseFusion (Submission #802)

Method	MPJPE	PA-MPJPE	PCK@.05	PCK@.10
A. Ours, calibrated ref., 52 joints, fold-0 val, paper protocol (m)				
best single view (right)	0.191	0.066	0.31	0.43
mean fusion	0.189	0.063	0.32	0.45
submitted full model	0.189	0.063	0.31	0.45
B. Ours, calibrated ref., 5-fold cross-subject, learned baselines of A3 (m)				
mean fusion	0.149	0.044	0.34	0.50
LT-style head, 1.9K	0.149	0.044	0.26	0.50
residual head, 8K	0.128	0.040	0.37	0.55
submitted arch. retrained, 870K	0.142	0.044	–	0.52
C. Drive&Act, official 3D ref., test subjects, zero-shot, 10 joints (mm): rigid† / PA				
mean fusion (best single view: 29.0 / 24.0)	31.6	23.1	–	–
submitted full model	32.5	23.7	–	–
DLT, official calibration (14 joints, absolute)	32.5	29.0	–	–

Table 1. Corrected results. †rotation+translation alignment only.

To all reviewers. Prompted by R-LAYQ's Tab. 3 anomaly, we audited the full evaluation stack and found four defects: (i) transposed rotation and norm-ratio scale in the Procrustes alignment; (ii) a zero-gradient gate-entropy term (clamp after log); (iii) a degenerate shoulder fallback collapsing whole frames; (iv) uncalibrated, constructed extrinsics in the triangulation reference (per-session scale error, median $2.4\times$). All are fixed and regression-tested. The reference is rebuilt by per-sequence bundle adjustment (reprojection $9.60 \to 3.31$ px); each sequence's free global scale is anchored to the SAM3D-predicted shoulder width (median 0.371 m; cross-sequence std $0.369 \to 0.015$ m); PA-MPJPE and reprojection are scale-independent. Tab. 1 gives all corrected numbers, one protocol and one valid-joint mask per block.

R-LAYQ W1 — metric anomaly. Cause: (i)+(iii). Synthetic check: the old code returns PA error 0.7–2.4 m where the truth is 0.04; the fix recovers it. Corrected, same masks: canonicalized mean fusion 0.971/0.564 m on the original reference — the reversal is gone; calibrated: Tab. 1A (the submitted model is fold-0 by the paper's protocol).

W2 — fallback rule. Correct inference. Missing rule: shoulder axis of the nearest earlier valid frame (a leading invalid prefix uses the first valid frame — the only lookahead); none valid $\to$ canonical $+x$ axis; a guard covers the axis-parallel case; pseudocode in §3.3.

W3 — independent reference. No independent MoCap subset was available; we no longer claim absolute 3D accuracy and instead bound the reference by validity/reprojection statistics (19jf W1) and reference sensitivity (conclusions unchanged across original/calibrated). Drive&Act (A2) serves only as cross-rig transfer evidence — its 3D reference is itself OpenPose-triangulated.

W4 — narrow the learned-component claims. We do: submitted model $\equiv$ mean fusion (Tab. 1A; see also A3). Claims narrow to canonicalization + uniform fusion; the learned parts' remaining value is corruption robustness (A1). Efficiency (0.87M params, 6.2 ms/frame CPU) is fusion-stage only; end-to-end latency incl. SAM3D is unmeasured. Minor: mapping table, frame-rate normalization, split/clip statistics, view-imbalance analysis added; joint-limit evaluation: future work.

R-19jf W1 — reference reliability under occlusion. Joints not triangulable from ≥ 2 views within the 40 px reprojection filter are invalid and excluded for *every* method via the same mask. Validity: head/shoulder–neck 1.00, hands 0.34; hands dominate the error (0.316 m vs. torso 0.067), so hands are reported separately, plus a torso-only protocol. Pipeline: manual checks on 2D inputs $\to$ automatic DLT $\to$ 40 px filter $\to$ per-sequence calibration, with per-sequence uncertainty statistics.

W2 — canonicalization mechanism. The gain is removal of per-view depth/scale inconsistency, not the fixed rig: canonicalized fusion beats every single view on our data (Tab. 1A) and, zero-shot, on Drive&Act (23.1 vs. 24.0–31.9 mm PA). (Minor) module necessity: see W4, A1.

R-xS2o A1 — gating collapse (W1–W2). We agree the stated entropy term, if active, favors uniform weights. Our audit found its implemented gradient exactly zero, so it did not cause the collapse; retraining without it, with the fix, and with a leave-one-view-out teacher still gave 0.3333 gates — the median pseudo-teacher supplies the dominant trivial optimum. We therefore remove the term, narrow the clean-input claim (W2 confirmed), and evaluate gating only under explicit view corruptions: corruption-augmented training makes a 1.9K gated head give a zeroed view weight 0.037, degrading +1–5% (mean fusion +18%; 0 for missing views).

A2 — Drive&Act (request 2). Protocol: official split-0 test subjects, official 3D reference, no Drive&Act training; the 10 joints shared with BODY-25; PA-MPJPE main (reference lacks hips for canonical alignment) and rigid-MPJPE, identical for all rows (Tab. 1C). We do not compare with published Drive&Act MPJPE (different joints/metrics). 0.412 m was the defect-(iv) artifact (W3); PCK@0.05/0.02 in Tab. 1 (M3).

A3 — deep-learning baselines (request 3). Learnable Triangulation (2D heatmaps + camera parameters) and Faster VoxelPose (feature volumes) are not input-matched to our 3D-keypoint-only interface (nor reproducible within the rebuttal). Input-matched references added: direct DLT (Tab. 1C; on our data it is the reference itself) and learned heads on identical input — confidence-weighted LT-style (1.9K), set-attention (28K), residual (8K), and the submitted architecture retrained under the same protocol (870K), which did not beat the simpler heads (Tab. 1B). Minor: SAM3D details added (M1); circularity (M2): calibration, A2.